

DSP EXPERIMENTS

EXP1.1

```
clc;
```

```
clear;
```

```
close all;
```

```
% Filter Specifications
```

```
Fsamp = 8000;    % Sampling frequency (Hz)
```

```
Fp = 1000;       % Passband frequency (Hz)
```

```
Fst = 3000;      % Stopband frequency (Hz)
```

```
Rp = 1;          % Passband ripple (dB)
```

```
Rs = 20;         % Stopband attenuation (dB)
```

```
% Normalize frequencies
```

```
Wp = Fp / (Fsamp/2);
```

```
Ws = Fst / (Fsamp/2);
```

```
% Calculate minimum filter order and cutoff frequency
```

```
[N, Wn] = buttord(Wp, Ws, Rp, Rs);
```

```
% Design Butterworth Low-Pass Filter
```

```
[b, a] = butter(N, Wn, 'low');
```

```
% Display filter coefficients
```

```
disp('Numerator coefficients (b):');
```

```
disp(b);
```

```
disp('Denominator coefficients (a):');
```

```
disp(a);
```

```
% Magnitude and Phase Response
```

```
figure;
```

```
freqz(b, a);
```

```
title('Magnitude and Phase Response');
```

```
% Impulse Response
```

```
figure;
```

```
[h, n] = impz(b, a, 50);
```

```
stem(n, h);
```

```
title('Impulse Response');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
% Step Response
```

```
figure;
```

```
stepz(b, a);
```

```
title('Step Response');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b, a);
```

```
title('Pole-Zero Plot');
```

```
grid on;
```

% Display filter order and cutoff frequency

```
fprintf('Minimum Filter Order (N) = %d\n', N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

```
fprintf('Cutoff Frequency in Hz = %.2f Hz\n', Wn * (Fsamp/2));
```



